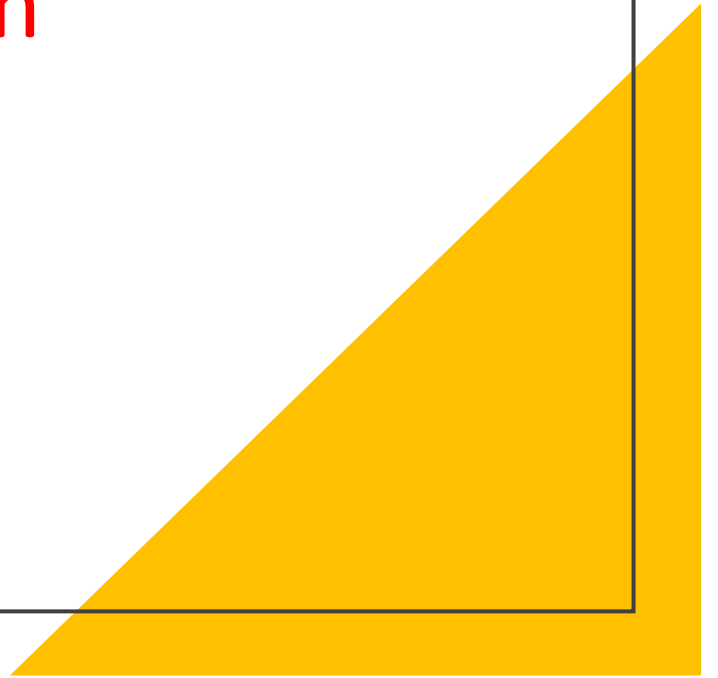


Overfitting Underfitting Regularization

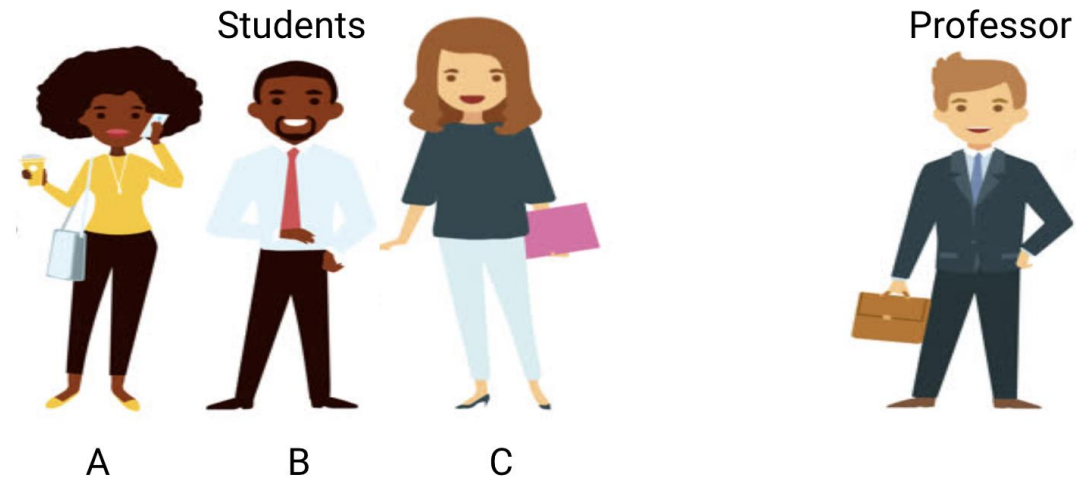


Model

- *In machine learning, generalization usually refers to the ability of an algorithm to be effective across a range of inputs and applications.*
- Suppose that we are designing a machine learning model.
- How a model is said to be a good machine learning model? -> By checking if it generalizes any new input data from the problem domain in a proper way.
- This helps us to make predictions in the future data, that data model has never seen.
- *A model is good if it neither Underfits or Overfits.*

Example to Understand Underfitting vs. Overfitting

- Consider a math class consisting of 3 students and a professor.
- Now, in any classroom, we can broadly divide the students into 3 categories.



- student A resembles a student who does not like math.
- She is not interested in what is being taught in the class and therefore does not pay much attention to the professor and the content he is teaching.



A

- Hobby = chating
- Not interested in class
- Doesn't pay much attention to professor

- student B. He is the most competitive student who focuses on memorizing each and every question being taught in class instead of focusing on the key concepts.
- Basically, he isn't interested in learning the problem-solving approach.



B

- Hobby = to be best in class.
- Mugs up everything professor says.
- Too much attention to the class work.

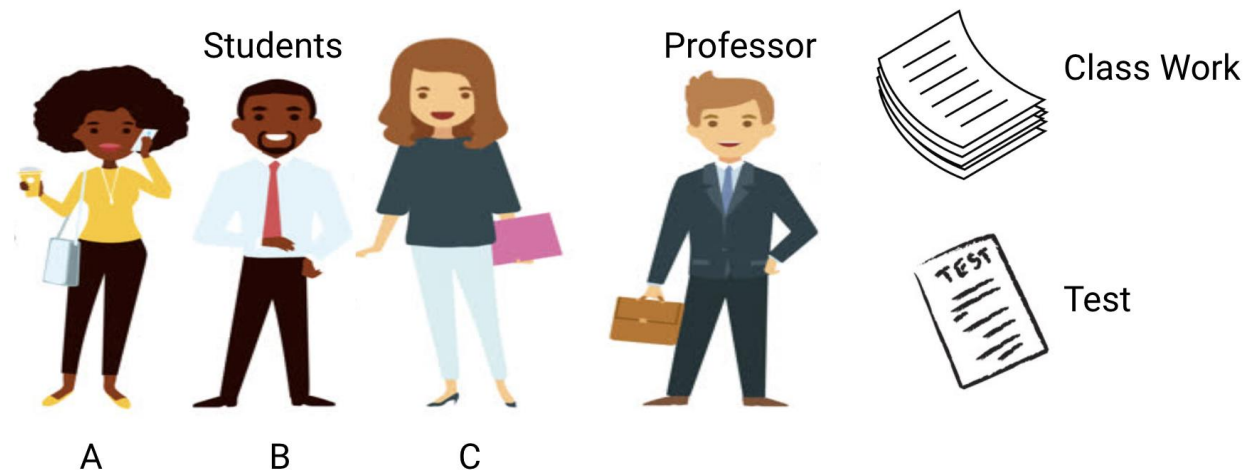
- Finally, we have the ideal student C. She is purely interested in learning the key concepts and the problem-solving approach in the math class rather than just memorizing the solutions presented.



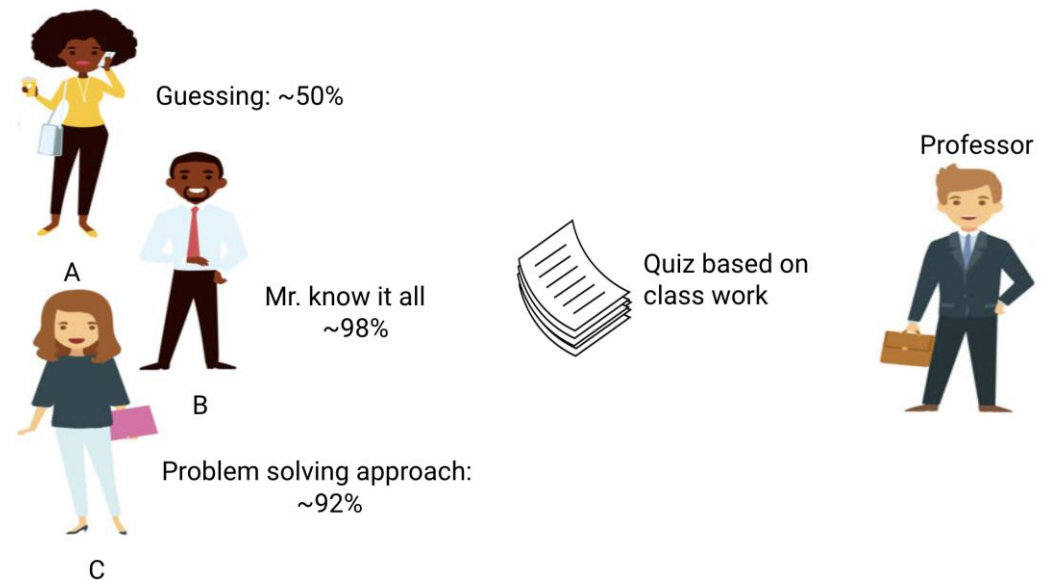
C

- Hobby = learning new things
- Eager to learn concepts.
- Pays attention to class and learns the idea behind solving a problem.

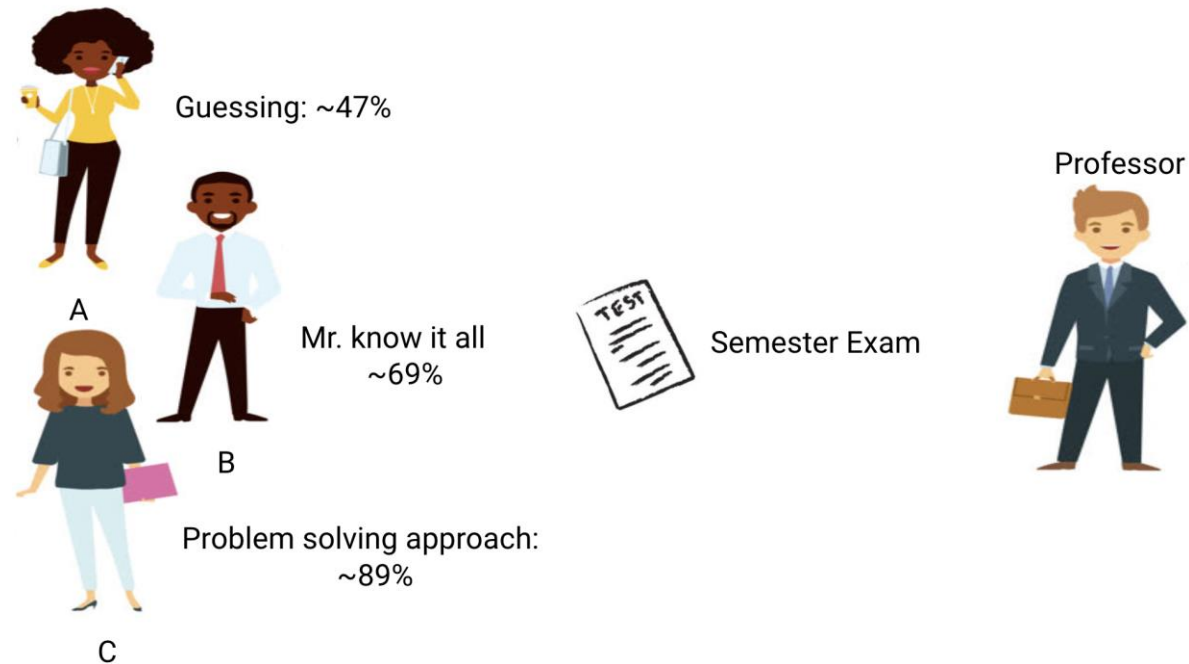
- The professor first delivers lectures and teaches the students about the problems and how to solve them. At the end of the day, the professor simply takes a quiz based on what he taught in the class.
- The obstacle comes in the semester tests that the school lays down. This is where new questions (unseen data) comes up. The students haven't seen these questions before and certainly haven't solved them in the classroom. Sounds familiar?

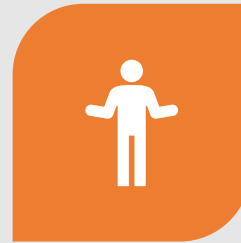


- Student A, who was distracted in his own world, simply guessed the answers and got approximately 50% marks in the test
- On the other hand, the student who memorized each and every question taught in the classroom was able to answer almost every question by memory and therefore obtained 98% marks in the class test
- For student C, she actually solved all the questions using the problem-solving approach she learned in the classroom and scored 92%



- Let's also look at what happens during the monthly test, when students have to face new unknown questions which are not taught in the class by the teacher.





IN THE CASE OF STUDENT A,
THINGS DID NOT CHANGE
MUCH AND HE STILL
RANDOMLY ANSWERS
QUESTIONS CORRECTLY ~50%
OF THE TIME.



IN THE CASE OF STUDENT B, HIS
SCORE DROPPED
SIGNIFICANTLY. CAN YOU GUESS
WHY? THIS IS BECAUSE HE
ALWAYS MEMORIZED THE
PROBLEMS THAT WERE TAUGHT
IN THE CLASS BUT THIS
MONTHLY TEST CONTAINED
QUESTIONS WHICH HE HAS
NEVER SEEN BEFORE.
THEREFORE, HIS PERFORMANCE
WENT DOWN SIGNIFICANTLY



IN THE CASE OF STUDENT C, THE
SCORE REMAINED MORE OR
LESS THE SAME. THIS IS
BECAUSE SHE FOCUSED ON
LEARNING THE PROBLEM-
SOLVING APPROACH AND
THEREFORE WAS ABLE TO
APPLY THE CONCEPTS SHE
LEARNED TO SOLVE THE
UNKNOWN QUESTIONS

How Does this Relate to Underfitting and Overfitting in Machine Learning?



A

Not interested in learning

Class test ~50%
Test ~47%



B

Memorizing the lessons

Class test ~98%
Test ~69%

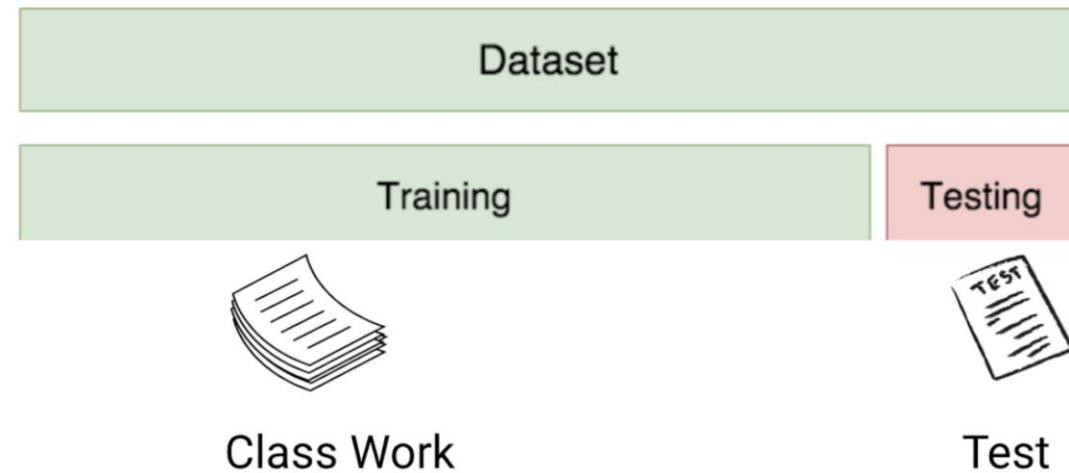


C

Conceptual Learning

Class test ~92%
Test ~89%

- First, the classwork and class test resemble the training data and the prediction over the training data itself respectively.
- On the other hand, the semester test represents the test set from our data which we keep aside before we train our model (or unseen data in a real-world machine learning project).





A

Not interested in learning

Class test ~50%

Test ~47%

Under-fit/ biased learning



B

Memorizing the lessons

Class test ~98%

Test ~69%

Over-fit/ Memorizing



C

Conceptual Learning

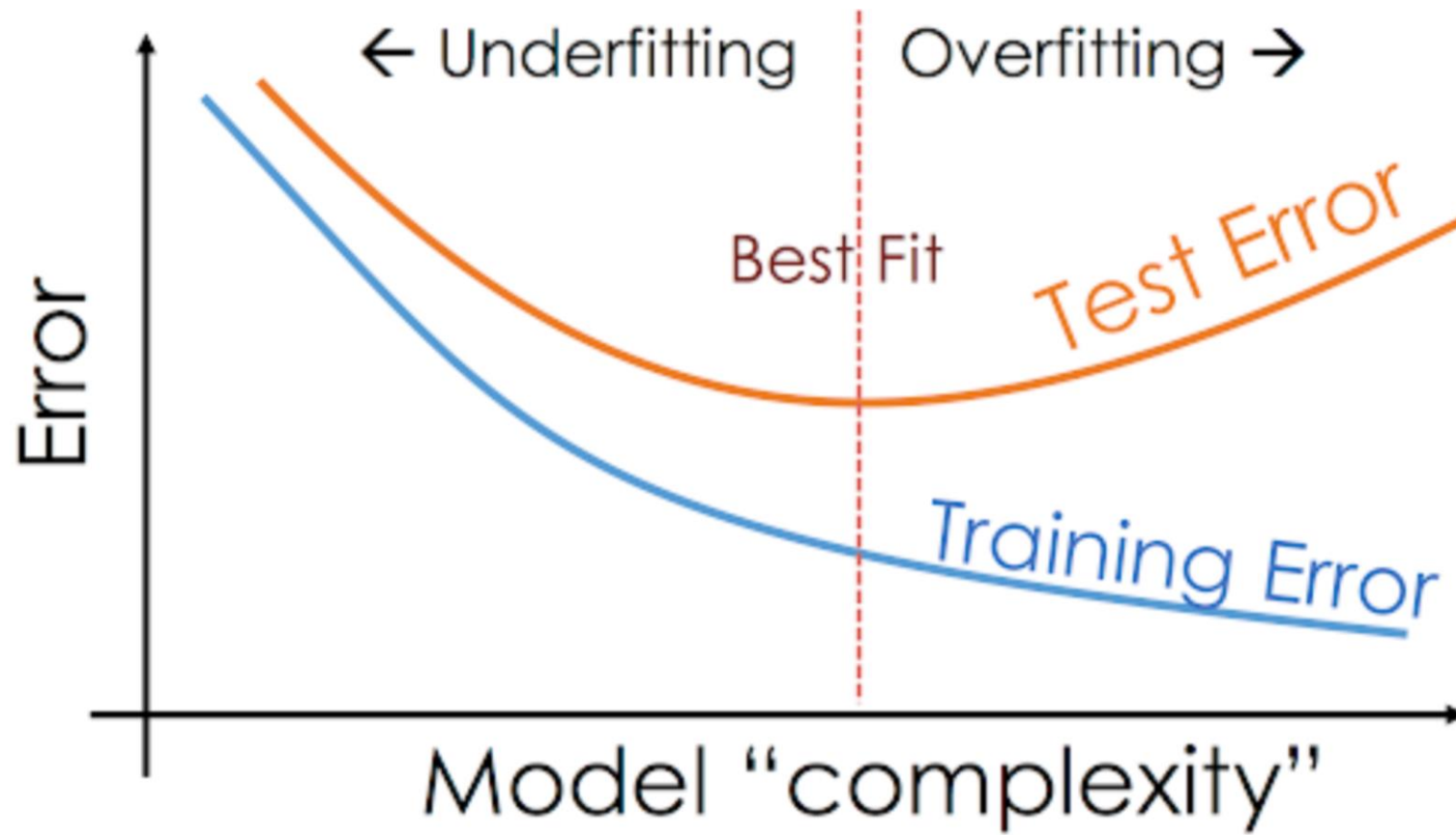
Class test ~92%

Test ~89%

Best-fit

“ *This situation where any given model is performing too well on the training data but the performance drops significantly over the test set is called an overfitting model.* ”

For example, non-parametric models like [decision trees](#), [KNN](#), and [other tree-based algorithms](#) are very prone to overfitting. These models can learn very complex relations which can result in overfitting. The graph below summarises this concept:



“ *On the other hand, if the model is performing poorly over the test and the train set, then we call that an underfitting model. An example of this situation would be building a linear regression model over non-linear data.*

Overfitting

- Can be generally termed as something when the ML model is extremely dependent on the training data.
- The model is build from each data point view of the training data that it is not able to learn and fails drastically when exposed to new test data.
- It mostly occurs in non-linear, non- parametric data.
- Non-parametric refers to an assumption that the data doesn't have a normal distribution.
- It occurs mostly in complex algorithms like tree based models, KNN etc.

Overfitting

- In simple words, Overfitting occurs when the model begins to memorize the training data instead of learning and understanding the underlying trend of the data. This results in the inefficiency of the model to perform well on new independent data.
- As we all know, an overfitting model learns the training data really well and close from each data point view that it fails when new data is presented to it
- A model with high variance and low bias is said to be overfitting.
- For eg: if the accuracy of a ML model is 99% on training data but 50% on test data, then it is a clear case of overfitting.

Overfitting can be overcome by :

- Cross validation techniques like K-fold cross validation.
- Boosting and bagging.
- Using linear models for linear data.
- Grid search techniques or including parameters like maximum depth (Number of trees) for tree based algorithms can go a long way in solving overfitting issues.
- Pruning in case of decision trees.
- L1 regularization (Lasso regularization/penalization).
- L2 regularization (Ridge regularization/penalization).

Underfitting:

- An underfitting model is neither good to the training data nor the test data.
- Inability of the model to understand/analyze the underlying trend/structure of the data, unable to derive critical information from the training data. This generally happens when there is not enough training data.
- **A model with high bias and low variance is said to be underfitting.**
- Underfitting might occur when we try to build linear models on non linear data.

Underfitting can be overcome by:

- Using feature selection, creating more features from existing features etc.
- Using Polynomial regression.
- For eg: if the accuracy of a ML model is 55% on training data and 50% on test data, then it is a clear case of underfitting because the model is not performing well on both train and test data.
- Underfitting mostly occurs in simple algorithms like Linear Regression